

CMPSC 465
Spring 2025
Final Exam

Name: _____

Time Limit: 110 Minutes

PSU Email ID: _____

This exam contains 14 pages (including this cover page, double-sided) and 4 questions.
Total of points is 76.

Instructions (must read)

- **Write your name on the top of each page.**
 - No cheat sheet or electronical devices are allowed in this exam.
 - No clarification question will be answered during exam. Please try your best to answer these questions.
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0. (0 points) (**Acknowledgement**) In this exam, you should write your solutions within the provided answer lines (for multiple choice questions) and solution boxes (for open-ended questions). Anything written outside these areas will not be graded no matter how correct they are.

(*Hint 1:* if you feel your solution is too long to fit in the provided solution box, then it is probably not be on the right track.)

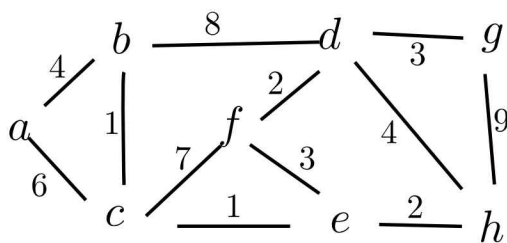
(*Hint 2:* if you feel your solution is too short for solution box, it's totally OK. Don't feel your solution is wrong because of its length. Some solutions can be described in two sentences, while I still give you a solution box of a full page.)

Write (copy) the following sentence in the following box. **Failed to do so will receive 0 for the final.**

"I acknowledge that I have to write my solutions within the provided answer lines and solution boxes. Anything written outside these dedicated areas will not be graded."

1. (24 points) Multiple choice (Exactly *one* choice is correct)

(1) (2 points) What is weight of a minimum spanning tree of the following graph?



- A. 15
- B. 16
- C. 17
- D. 14

(1) **B**

- (2) (2 points) Consider a weighted and undirected graph with positive weights. If the weight of each edge is decreased by a positive amount d , which of the following is true?

- A. The minimum spanning tree won't change only if $d = 1$.
- B. The shortest path could change only if $d = 1$.
- C. If the weights remain positive, the minimum spanning tree won't change, but the shortest path might change.
- D. If the weights remain positive, both the minimum spanning tree and the shortest path won't change.

(2) **C**

- (3) (2 points) Which of the following codes is possibly generated by Huffman's algorithm

- A. $\{1, 0, 10\}$
- B. $\{0, 100, 101, 110, 111\}$
- C. $\{100, 101, 0\}$
- D. $\{00, 01, 10\}$

(3) **B**

- (4) (2 points) Consider two strings "TACGCA" and "ATGCC". Which of the following is true?

- A. The alignment with the minimum edit distance is unique.
- B. The minimum edit distance is 4.
- C. The minimum edit distance is 5.
- D. The minimum edit distance is 6.

(4) _____ **A** _____

(5) (2 points) Which of the following statements about the 0-1 knapsack problem is true?

- A. The greedy algorithm we saw in the lecture returns an optimal solution.
- B. The greedy algorithm we saw in the lecture returns a solution with a bounded approximation ratio.
- C. The dynamic programming algorithm we learned in the lecture returns an optimal solution.
- D. The dynamic programming algorithm we learned in the lecture runs in polynomial time.

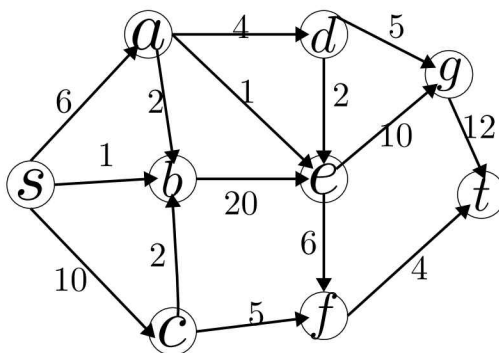
(5) _____ **C** _____

(6) (2 points) Which of the following statements about dynamic programming is true?

- A. Dynamic programming algorithms are always efficient (running in polynomial time).
- B. If a dynamic programming algorithm fills in an M -by- N table, then its running time is $O(MN)$.
- C. In dynamic programming, each problem is solved only once.
- D. If a problem can be solved by dynamic programming, it can also be solved using divide and conquer.

(6) _____ **C** _____

(7) (2 points) For the following flow network, with edge capacities as shown. What is the capacity of the minimum s - t cut?



- A. 13
- B. 12
- C. 11
- D. 14

(7) _____ **A** _____

(8) (2 points) Given a flow network, if we add 1 to the capacity of every edge, which of the following is true?

- A. The value of the maximum flow will be increased by at least 1.
- B. It is possible that the value of the maximum flow will remain the same.
- C. The capacity of the minimum cut will be decreased by at least 1.
- D. The value of the maximum flow will be increased by 1.

(8) _____ **A** _____

- (9) (2 points) Suppose you want to maximize the function $f(x_1, x_2) = x_1 \cdot x_2 - \frac{1}{2}x_1 - \frac{1}{2}x_2$ on the regions $0 \leq x_1 \leq 10$ and $0 \leq x_2 \leq 15$. Which of the following is true?

- A. The simplex algorithm can be used to solve this problem.
- B. The feasible region is nonlinear.
- C. The objective function is nonlinear.
- D. This formulation is in the standard form of a linear programming formulation.

(9) _____ **C** _____

- (10) (2 points) Which of the following statements about linear programming is true?

- A. If the feasible region is unbounded, then there is no solution.
- B. Equality constraint is not allowed.
- C. Any primal feasible solution is strictly larger than any dual solution.
- D. Any dual feasible solution is an upper bound on any primal feasible solution.

(10) _____ **D** _____

- (11) (2 points) Let X and Y be two problems. If X is polynomial-time reducible to Y , which of the following is true?

- A. If Y can be solved in polynomial time, then X can be solved in polynomial time.
- B. If Y cannot be solved in polynomial time, then X cannot be solved in polynomial time.
- C. X is harder than Y .
- D. If X can be solved in polynomial time, then Y can be solved in polynomial time.

(11) _____ **A** _____

- (12) (2 points) Which of the following statements is false?

- A. The class **NP-complete** contains the hardest problems in **NP**.
- B. It is known that **NP-complete** problems cannot be solved in polynomial time.
- C. All problems in **P** have an efficient certifier.
- D. All problems in **NP** have an efficient certifier.

(12) _____ **B** _____

2. (24 points) A thunderstorm just hit Centre County and caused massive power outage, including the Downtown area in State College. The city government decided to place a number power generators along East College Avenue to provide powers to the restaurants there. There are certain restrictions on placing these generators: if a generator is placed on the i -th block, then no generator shall be placed on its adjacent four blocks, i.e., the $(i - 2)$ -th, $(i - 1)$ -th, $(i + 1)$ -th, and $(i + 2)$ -th blocks due to noise regulations.

Suppose the city government hired you to place these generators so that some restaurants get power and then can serve customers. The objective is to maximize the revenue of these restaurants in operation. According to the city tax records, the revenue generated by restaurants in the i -th block is r_i per day. Now, you need to design a dynamic programming algorithm to find out an optimal placement of generators for the maximum revenue.

Suppose there are n blocks on East College Avenue. To begin with, let us define the subproblem, $G(j)$, as the maximum revenue generated per day if the generators can only be placed among blocks $1, 2, \dots, j$.

- (1) (2 points) What is the maximum total revenue? Answer this question using the subproblem defined above.

Solution: $G(n)$

- (2) (5 points) What is the recurrence relation for $G(j)$?

Solution:

$$G(j) = \max\{G(j-1), r_j + G(j-3)\}$$

- (3) (5 points) Specify the base case.

Solution: $G(1) = r_1$, $G(2) = \max\{r_1, r_2\}$. We can also add other base cases such as $G(0) = 0$ and $G(3) = \max\{r_1, r_2, r_3\}$. But $G(1)$ and $G(2)$ must be here.

- (4) (5 points) Define an auxiliary table that helps you construct the optimal locations to place generators.

Solution:

$$\text{prev}(j) = \begin{cases} j - 3, & \text{if } G(j) = r_i + G(j - 3) \\ j - 1, & \text{Otherwise} \end{cases}$$

- (5) (5 points) Explain how to obtain the optimal locations using the auxiliary table defined above.

Solution: We first need to check if a generator will be placed on the n -th block by checking if $G(n) = r_n + G(j - 3)$. Then we start with $\text{prev}(n)$ and do a back tracking (i.e., if we check $\text{prev}(j)$, we set $j' = \text{prev}(j)$ and next check $\text{prev}(j')$. If the last entry is 1, then we place a generator on the first block; otherwise (it is 2), we need to decide whether to place a generator on the first or second block by checking if $G(2) = r_1$.

Note that it's also correct to define a True/False table in Part (4), but we still need to back track in this case.

- (6) (2 points) What is the running time of this dynamic programming algorithm? Briefly justify your answer.

Solution: $O(n)$, since there are $O(n)$ entries to compute and computing each entry takes $O(1)$ time.

3. (28 points) The thunderstorm not only destroyed power, it also interrupted network connections within campus. To help n students prepare their exams, the university IT department decides to place n portable Wi-Fi routers around the campus. Let $b_1, b_2, \dots, b_n$ be the locations of n portable Wi-Fi routers. Also let $s_1, \dots, s_n$ be the locations of these students. Due to budget constraints, these Wi-Fi routers are pretty weak: each of them can connect to *only one* device, and can service a distance at most Δ .
- (1) (8 points) Describe an algorithm that can quickly determine whether or not all students can get connected to Wi-Fi, given the location $b_1, \dots, b_n, s_1, \dots, s_n$, and the parameters Δ . (*Hint:* try to reduce this problem to the bipartite matching problem.)

Solution: Construct a bipartite graph $G = (V, E)$, where $V = \{b_1, \dots, b_n\} \cup \{s_1, \dots, s_n\}$, and there's an edge between b_i and s_j when the distance between them is no larger than Δ . All students get connected if and only if there's a bipartite matching in G of size n .

- (2) (4 points) What is the running time of your algorithm? Justify your answer.

Solution: The bipartite graph G has $2n$ vertices and $O(n^2)$ edges. The bipartite matching algorithm in the lecture runs in time $O(|V||E|)$. So the running time is $O(n^3)$.

- (3) (8 points) Now, let's take a digression to review a flow problem. Suppose you are given a directed graph $G = (V, E)$ with positive integer capacity c_e on each edge e , a source $s \in V$ and a sink $t \in V$. You are also given a maximum flow f in G . The flow is *acyclic*: there are no cycles in G on which all edges carry positive flow. The flow f is also integer valued.

In the homework, you showed that if the capacity of an edge $e^* \in E$ is reduced by 1, a maximum flow in the new flow network can be found in time $O(|E| + |V|)$.

Now, suppose the capacity of an edge e^* is *increased* by 1. Show that a maximum flow in the new flow network can still be found in time $O(|E| + |V|)$.

Solution: Assume the flow f is integer-valued. Let $e^* = (v, w)$. If $f(e^*) < c_{e^*}$, then increasing its capacity by 1 does not cause a problem.

Otherwise, if $f(e^*) = c_{e^*}$, increasing the capacity of e^* by 1 doesn't change the feasibility of the f , i.e., f is still a feasible flow. Then, we just build the residual graph G_f and try to find an augmenting path, and then update the flow if an augmenting path can be found.

Building G_f takes $O(|E| + |V|)$ time and find a path can be done using BFS. Hence, the total cost is $O(|E| + |V|)$.

- (4) (8 points) Back to the connectivity problem. Suppose all students were connected. But now one student moved to a new location. Describe an algorithm, that given the new locations of all students, quickly determine whether or not all students can still get connected without recomputing from scratch. Aim for an algorithm with running time $O(n^2)$, which is faster than re-applying the algorithm in Part (1) from scratch. (*Hint:* the results in Part (3) should help.)

Solution: Once a student moved to a new location, we update the graph G built in Part (1), by removing and adding edges if necessary. Among the removed edges, at most one of them carried flow. And in the new graph, the size of the new matching cannot exceed the size of the old matching. So the size of the new matching is either unchanged, or reduced by 1. (Implied by Part (3)).

Then, we solve the bipartite matching problem, which further reduced to the maximum matching problem with edge capacity 1. We just run one iteration of Ford-Fulkerson (or equivalently, updating the residual graph, finding an s - t path in the update residual graph), a new max flow can be found, which determined the size of the matching, which then further determines if all students can get their preferred room. Everything can be done in $O(|V| + |E|) = O(n^2)$ time.